

Module 1

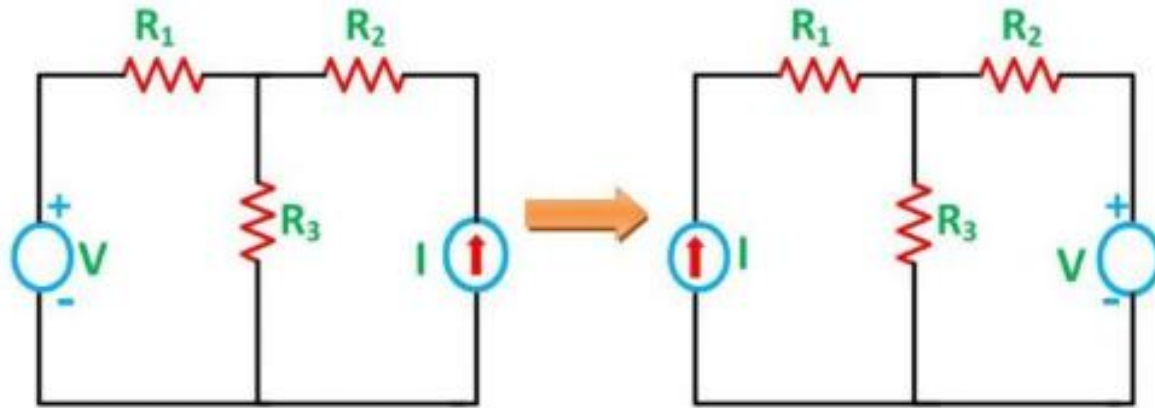
DC Circuit Analysis and Theorems

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Reciprocity Theorem

The reciprocity theorem in electrical circuits states that in a linear, bilateral network with a single voltage source, the ratio of the voltage across one element to the current through another element remains the same even if the voltage source and the current measurement point are interchanged.



Resistances R_1 , R_2 , R_3 are connected in the circuit with a voltage source (V) and a current source (I).

The voltage source and current source can be interchanged for solving the network with the help of Reciprocity Theorem.

Reciprocity Theorem

The limitation of this theorem:

It is applicable only to single-source networks and not in the multi-source network.

The network where reciprocity theorem is applied should be linear and consist of resistors, inductors, capacitors and coupled circuits.

The circuit should not have any time-varying elements.

Reciprocity Theorem

Steps for solving a network utilizing Reciprocity Theorem:

Step 1 – Firstly, select the branches between which reciprocity has to be established.

Step 2 – The current in the branch is obtained using any conventional network analysis method.

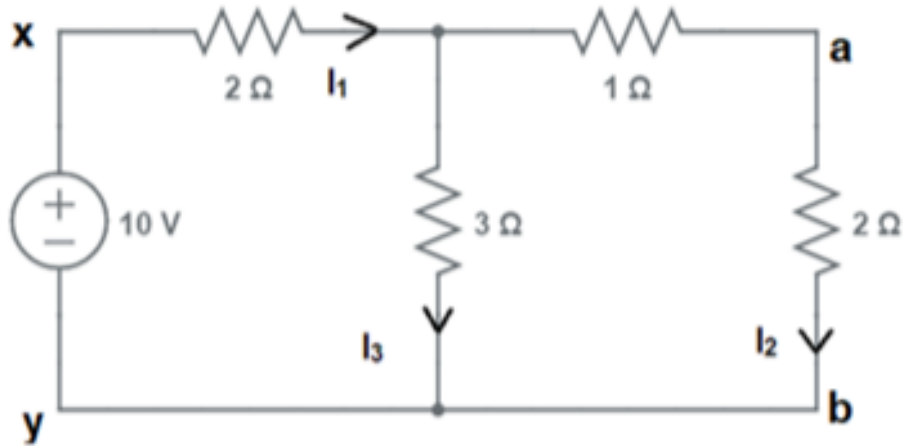
Step 3 – The voltage source is interchanged between the branch which is selected.

Step 4 – The current in the branch where the voltage source was existing earlier is calculated.

Step 5 – Now, it is seen that the current obtained in the previous connection, i.e., in step 2 and the current which is calculated when the source is interchanged, i.e., in step 4 are identical to each other.

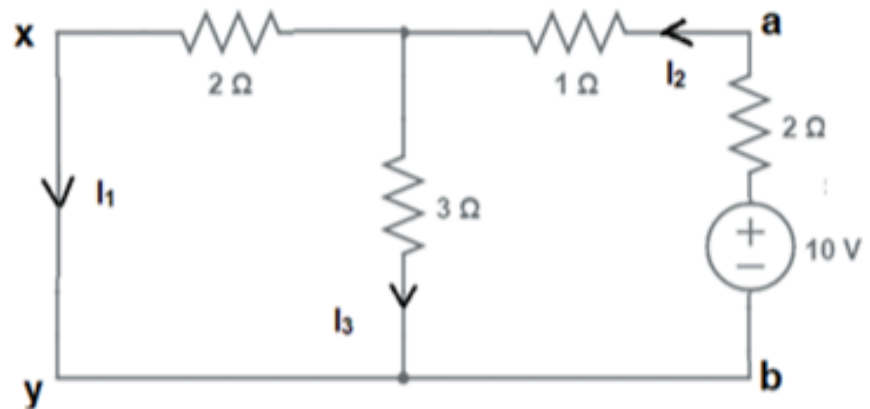
Reciprocity Theorem

Test the validity of the reciprocity theorem in branches x-y and a-b.



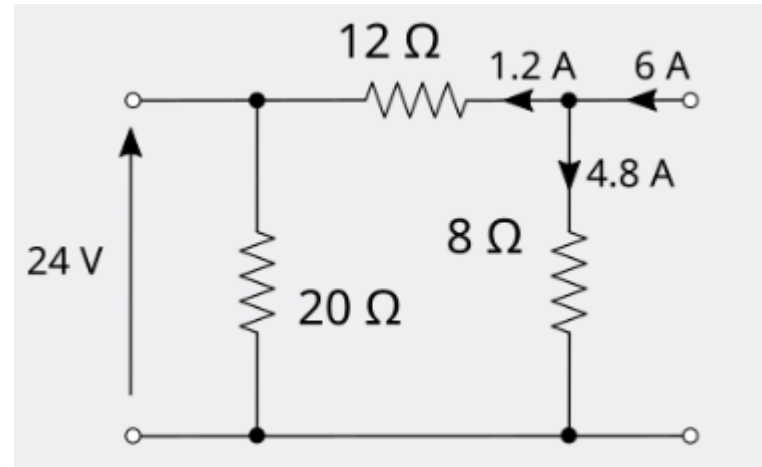
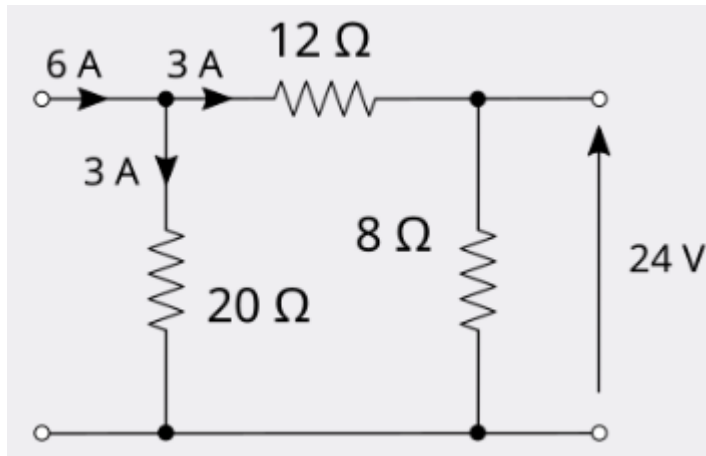
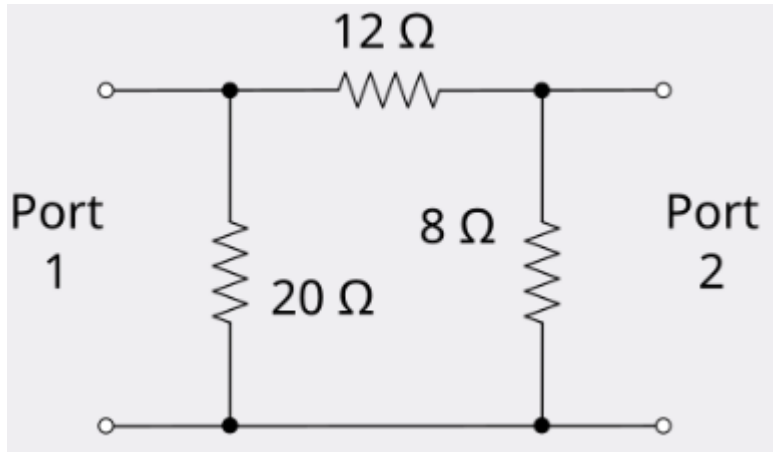
$$I_1 = 1.4 \text{ A}$$

$$I_2 = 1.4 \text{ A}$$



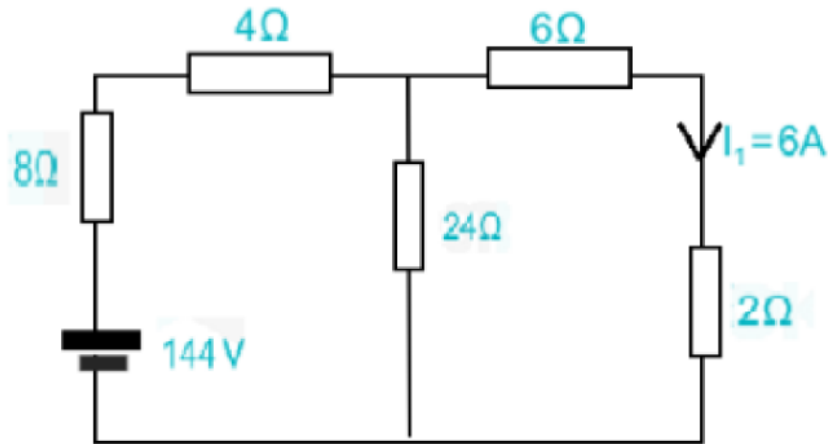
Reciprocity Theorem

Test the validity of the reciprocity theorem in the network if 6A is injected in port 1 and 24 V is measured at port 2.

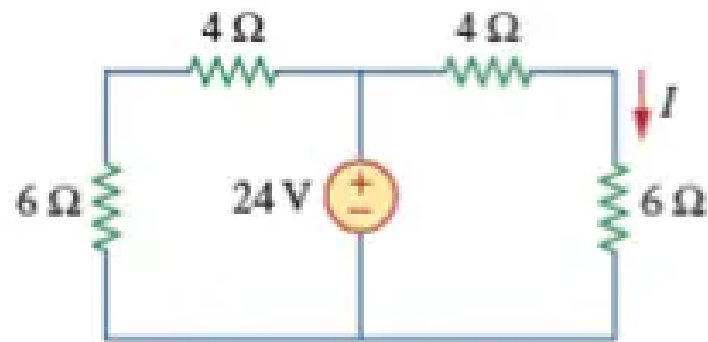


Reciprocity Theorem

Validate the reciprocity theorem.



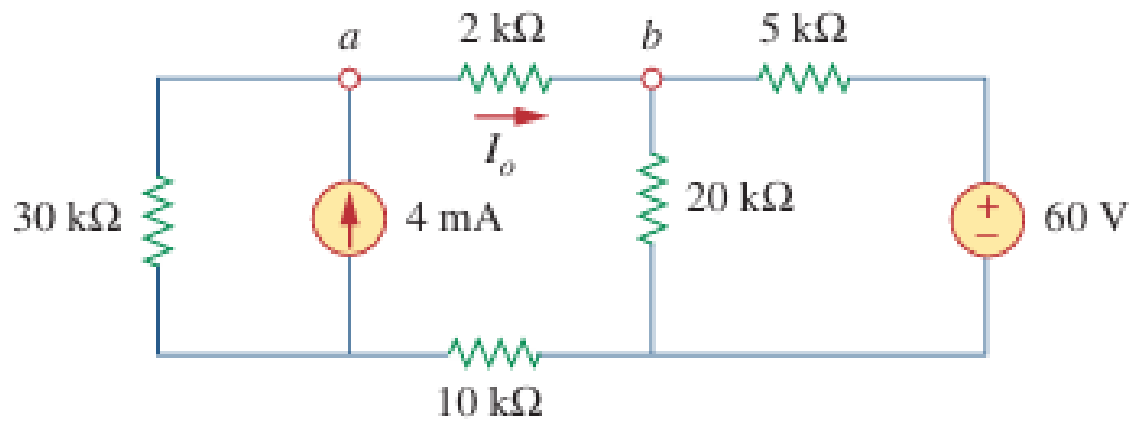
(a)



(b)

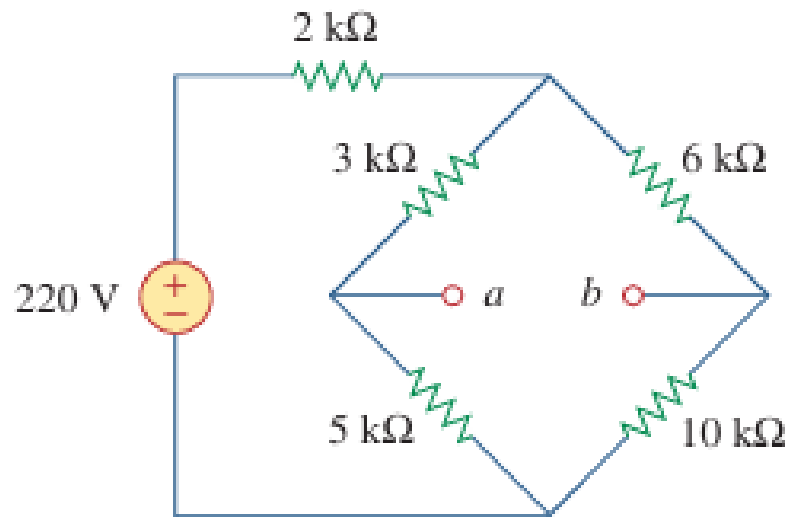
Exercise

Consider the circuit given below. An ammeter with internal resistance is inserted between A and B to measure I_o . Determine the reading of the ammeter if: (a) $R_i = 500 \, \Omega$ (b) $R_i = 0 \, \Omega$ (Hint: Find the Thevenin's equivalent circuit at terminals.



Exercise

Consider the bridge circuit below. Is the bridge balanced? If the 10-k resistor is replaced by an 18-k resistor, what resistor connected between terminals absorbs the maximum power? What is this power?



Next class: Tutorial-Problem solving